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Additive Manufacturing of Topology-Optimized Graded Porous Structures: An Experimental Study

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Graded porous structures, combining the high stiffness of bulk designs and the robustness of porous structures, have received increasing attention in the fields of topology optimization and additive manufacturing. This study aims to experimentally investigate the properties of topology-optimized and additively manufactured graded porous structures, with comparisons to conventional bulk designs and one-level porous structures. Examples with graded porosity are designed through a unique multiporosity topology optimization framework. This multiporosity framework generalizes the concept of multiple materials; i.e., each material field has a different level of local porosity, thus realizing the automatic distribution of multilevel porosity. The optimized designs are fabricated via the mask stereolithography process using photosensitive resins. Based on three-point bending tests, we study the failure processes and the influences of material deficiency on elastic stiffness for the three types of optimized designs: bulk, one-level porous, and graded porous designs. Experimental results demonstrate that the additively manufactured optimized graded porous structures not only have relatively high structural stiffness and load-carrying capacities but also show a ductile failure mode and certain robustness against material deficiency. The presented work contributes to experimental studies on demonstrating the combined advantages of topology-optimized graded porous structures by comparing the structural performances of graded porous structures with bulk designs and one-level porous structures.

INTRODUCTION

Topology optimization is a computational method for finding the most efficient material layouts within a given design domain, by optimizing an objective while fulfilling a single or multiple constraints. While topology-optimized designs can contain intricate geometries, additive manufacturing offers a promising avenue to fabricate optimized designs with complex geometries for demonstrations, experiments, and real-world applications. In the fields of topology optimization and additive manufacturing, the design of graded porous structures has become an emerging area of research because they combine the advantages of porous structures, which are robust with respect to material deficiency, force variation, and buckling,[1–3] and bulk designs, which have relatively high structural stiffness and load capacity.

Various topology optimization approaches have been proposed to design structures with one or multiple levels of porosities, such as the density-based formulation with local constraints,[1,2,4] the homogenization-based optimization method,[5,6] multiscale topology optimization frameworks,[7,8] and the moving morphable components and moving morphable voids approach.[9] In terms of experimental studies on topology-optimized structures with multiple levels of porosities, Clausen et al.[10] optimized shell-infill structures via a density-based topology

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optimization framework with a coating approach and experimentally demonstrated improved buckling performance compared with the conventional bulk design. Cheng et al.[11] designed functionally graded lattice structures using an asymptotic homogenization-based topology optimization method and validated the improved elastoplastic properties (e.g., stiffness and yielding load) of the optimized graded lattice design compared with a uniform lattice beam. However, few studies have experimentally investigated failure and fail-safe behaviors, specifically the structural failure modes and elastic stiffness with respect to material deficiency, of topology-optimized graded porous structures through comparisons with both bulk designs and one-level porous structures.

To fill the gap of few experimental studies of graded porous structures designed by topology optimization, we aim to first optimize novel systems with multiple levels of porosities (i.e., graded porous structures) as well as bulk designs and structures with one level of porosity through a unique multiporosity topology optimization framework.[12] Built upon the local volume constraint for one-level porosity,[1] this framework adopts a multiporosity local volume constraint, enabling smooth transitions among different levels of porosities. We then use photopolymer resins to fabricate the optimized designs using the mask stereolithography (MSLA) additive manufacturing process, and conduct three-point bending tests to experimentally investigate several structural characteristics, i.e., the structural stiffness, failure processes, and fail-safe behaviors, of those three types of structures. Therefore, this work contributes to experimental studies on exploring the advantages of topology-optimized graded porous structures by comparatively demonstrating the structural performance of bulk designs, one-level porous structures, and graded porous structures. The remainder of this paper is organized as follows: Sect. 2 presents the adopted topology optimization approach, optimized designs, manufacturing techniques, and experimental setups. Section 3 demonstrates and discusses the experimental results. Section 4 summarizes the observations and discusses limitations and future directions of this study.

METHODOLOGY, NUMERICAL RESULTS, AND ADDITIVE MANUFACTURING

This section briefly presents the multiporosity topology optimization approach that is capable of generating optimized graded porous structures, followed by demonstrations of optimized results. In addition, we present the fabrication techniques and experiment setups.

Topology Optimization Framework for Graded Porous Structure Design

We employ the multiporosity topology optimization framework[12] to generate bulk designs, one-level porous structures, and graded porous structures with the assumptions of linear elasticity and small deformations. This novel framework generalizes the concept of multiple materials; i.e., each material field can have not only distinct mechanical properties but also different levels of porosity (or a combination of both). We separate the density and the porosity design variables so as to parameterize the graded porous structures. The density design variable (denoted by $\boldsymbol{\rho}$) is filtered and projected to the physical density field (denoted by $\overline{\boldsymbol{\rho}}$), which represents the presence of materials in the design. The porosity design variable (denoted by $\boldsymbol{\xi}$) is filtered, projected, and mapped to the physical porosity fields (denoted by $\boldsymbol{m}$), which characterize multiple levels of porosity. We present the optimization formulation as follows:

$$
\begin{aligned}
\min_{\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)}} \; & C(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)},\boldsymbol{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)})) \\
\text{s.t. } & 0<\rho_e\le 1, \quad e=1,\ldots,N_e, \\
& -1\le \xi_e^{(k)}\le 1, \quad e=1,\ldots,N_e, \quad k=1,\ldots,N_\xi, \\
& G_G^{(j)}(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)})\le 0, \qquad j=1,\ldots,N_m, \\
& G_L(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)})\le 0, \\
& \text{with} \quad \boldsymbol{K}\boldsymbol{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}^{(1)},\ldots,\boldsymbol{\xi}^{(N_\xi)})=\boldsymbol{F},
\end{aligned}
\tag{1}
$$

where N_e is the number of elements, N_ξ is the number of porosity design variables, and N_m is the number of global volume constraints (i.e., the number of porosity levels). In the nested equilibrium equation, $\boldsymbol{K}$ is the global structural stiffness matrix, $\boldsymbol{F}$ is the external loading, and $\boldsymbol{u}_h$ is the equilibrating displacement field. The objective $C(\cdot)$ is the structural compliance. Note that, although we experimentally investigate structural characteristics such as failure modes and fail-safe behaviors in Sect. 3, the objective in the design formulation is to maximize the structural stiffness without mathematically accounting for robustness or fracture in the formulation. The constraint $G_G^{(j)}(\cdot)$ is to restrict the global volume for the jth level of porosity with an upper bound of $V_0^{(j)}$, and $G_L(\cdot)$ is the multiporosity local volume constraint, which drives the appearance of different levels of porosities. Note that, due to the objective of compliance minimization, the least porosity (providing the largest structural stiffness) is the prior choice of the optimizer; thus, the individual global volume constraints are required to ensure all the prescribed porosities to appear in the results.

In formulation (1), to avoid checkerboard and mesh-dependency issues, both $\boldsymbol{\rho}$ and $\boldsymbol{\xi}^{(k)}$ are first

filtered to $\widetilde{\boldsymbol{\rho}}$ and $\widetilde{\boldsymbol{\xi}}^{(k)}$, respectively, using a density filter[13] with the same radius r. Then, to ensure black-and-white designs, the filtered variables $\widetilde{\boldsymbol{\rho}}$ and $\widetilde{\boldsymbol{\xi}}^{(k)}$ are projected to the physical density field $\overline{\boldsymbol{\rho}}$ and the projected material/porosity variable $\overline{\boldsymbol{\xi}}^{(k)}$ using the Heaviside function[14] with sharpness parameters β_ρ and β_ξ, respectively. Finally, we apply the bi-value coding parameterization (BCP) approach[15] to map from the projected porosity variable $\overline{\boldsymbol{\xi}}^{(k)}$ to the physical porosity field $\boldsymbol{m}^{(j)}$. We perform the interpolation of Young's modulus with a modified solid isotropic material with penalization (SIMP)-like scheme[13,16] using density and material penalization parameters of p_ρ and p_m, respectively. The expression is given by

$$\hat{E}_e = \epsilon_0 + (1-\epsilon_0)\overline{\rho}_e^{p_\rho}\sum_{j=1}^{N_m}(m_e^{(j)})^{p_m}E^{(j)}, \tag{2}$$

where $\hat{E}_e$ is the interpolated Young's modulus for element e, ϵ_0 is a sufficiently small value to avoid numerical instability, and $E^{(j)}$ is the Young's modulus for the jth level of porosity. In this study, we experimentally investigate single-material designs with multiple porosity levels and assign $E^{(j)} = E, j = 1, \ldots, N_m$.

The key component in the formulation for generating graded porosity is the multiporosity local volume constraint.[12] The local volume constraint (G_L) takes the following form:

$$G_L(\boldsymbol{\rho}, \boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(N_\xi)}) = \left(\frac{1}{N_e}\sum_{e=1}^{N_e}\left(\frac{\hat{\rho}_e(\boldsymbol{\rho})}{\sum_{j=1}^{N_m} m_e^{(j)}(\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(N_\xi)})\alpha_0^{(j)}}\right)^q\right)^{\frac{1}{q}} - 1 \le 0, \tag{3}$$

where $\alpha_0^{(j)}$ is the upper bound of the local volume fraction within a circular influence region, which can be written as $\mathcal{I}(R)$ with a radius of influence of R. In the constraint, the local volume is defined with respect to the density field (i.e., $\hat{\rho}_e = \frac{\sum_{i\in\mathcal{I}_e(R)}\overline{\rho}_i A_i}{\sum_{i\in\mathcal{I}_e(R)} A_i}$, where A_i is the area of element i). To constrain the local volume fraction over the entire design domain, we apply the p-norm approximation with the parameter q. Note that, if only one porosity is assigned in the formulation, the multiporosity local volume constraint will reduce to the local volume constraint proposed in Ref. 1. Also, the framework does not require a predefined spatial distribution of multiple levels of porosities (see, e.g., Ref. 2 for a study that can predefine spatial distributions of porosity) and the optimizer distributes different levels of porosities to maximize the structural stiffness. For details of the multiporosity local volume constraint and the formulation, the reader is referred to Ref. 12.

Topology-Optimized Designs: Bulk, One-Level Porous, and Graded Porous Structures

This subsection presents the optimized designs, including the design domains, boundary conditions, parameter setups, and optimized topologies. We optimize a Messerschmitt–Bölkow–Blohm (MBB) beam and a curved beam for the failure test and the fail-safe test under material deficiency, respectively. For all the designs, we choose a relatively small value for the p-norm parameter ($q = 6$) to reduce weak joints at crossing members.[2]

MBB Beam

The purpose of the MBB beam example is to test and compare the failure processes of the bulk design, the one-level porous structure, and the graded porous structure. Employing the structural symmetry, half of the design domain (discretized by 67,500 structured quadrilateral elements) is

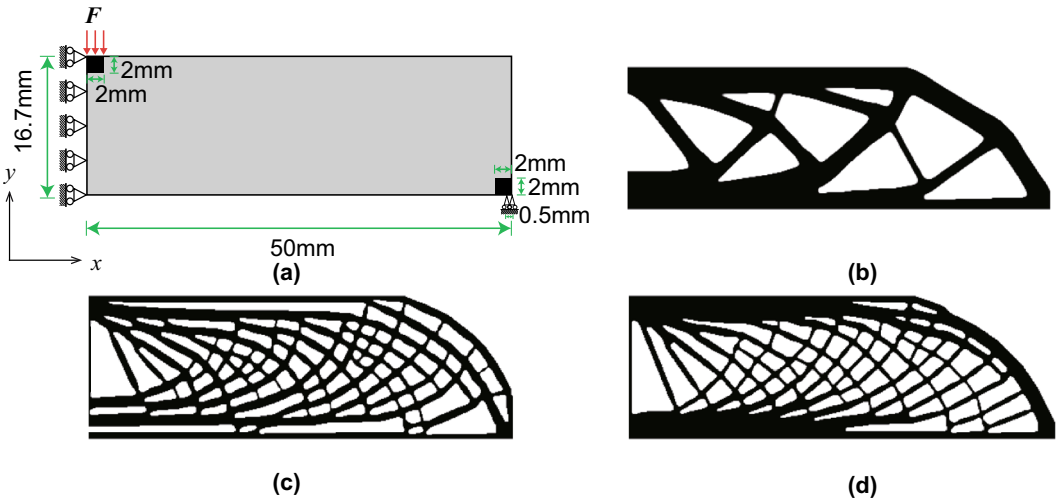


Fig. 1. Topology optimization of MBB beam: (a) design domain and boundary conditions, (b) optimized bulk design, (c) optimized one-level porous structure, and (d) optimized graded porous structure.

optimized (Fig. 1a). We set two passive zones at the top left and bottom right to increase contact areas and thus avoid possible damage to thin members caused by rotations around the roller supports and load cells during the experiment. For the bulk design, $r = 1.5$ mm; for the one-level porous structure, $\alpha_0 = 0.6$, $r = 0.8$ mm, and $R = 1$ mm; for the graded porous structure, $\alpha_0^{(1,2,3)} = [1, 0.7, 0.4]$, $r = 0.8$ mm, and $R = 1$ mm. It is ensured that the three optimized designs have the same volume fraction of 0.5. The optimized topologies shown in Fig. 1b–d reveal that the graded porous structure combines the features of the bulk designs and one-level porous structures—the outer regions contain bulk members, while the inner region contains two different levels of porosity with smooth transitions.

Curved Beam

We aim to use a curved beam example to measure and compare the elastic stiffness of the bulk design, the porous structure, and the graded porous structure with respect to material deficiency. Similar to the MBB beam design, the structural symmetry and the passive zones are imposed. The design domain (discretized by 94,244 structured quadrilateral elements) and boundary conditions are shown in Fig. 2a. For the bulk design, $r = 1.575$ mm; for the one-level porous structure, $\alpha_0 = 0.59$, $r = 1.05$ mm, and $R = 1.26$ mm; for the graded porous structure, $\alpha_0^{(1,2,3)} = [1, 0.7, 0.4]$, $r = 1.05$ mm, and $R = 1.26$ mm. It is ensure that the three optimized designs have the same volume fraction of 0.49. As shown in Fig. 1b–d, the optimized graded porous design achieves the automatic distribution of multilevel porosity, which combines the design features of both the bulk and one-level porous structures. To demonstrate a certain degree of structural robustness and fail-safe behavior, we evaluate the amount of elastic stiffness reduction under material deficiency at different locations. To do so, we apply a simplified local damage model[1,17] on the optimized structures to simulate a type of material defect. As demonstrated in Fig. 3, we remove material from four prescribed locations (defined by a circle with radius of 4.75 mm) to investigate the influence of defect locations on structural stiffness. Note that the size of the defects is strategically chosen to match the member size of the bulk design.

Additive Manufacturing of Optimized Designs and Experimental Setup

The specimens were fabricated through the MSLA process using the Prusa SL1 3D printer with a liquid-crystal display panel and an ultraviolet light-emitting diode panel. To generate 3D-printable models, we first extruded the optimized designs from 2D to 3D with a thickness of 8 mm and 10.5 mm for the MBB beam and the curved beam, respectively, then postprocessed the extruded results to .stl files using TopSlicer.[18] The .stl files were converted to .sl1 files for the SL1 printer. After printing, we used the Prusa CW1 machine to wash, dry, and cure the specimens. Figure 5a, b shows two examples (a graded porous MBB beam and a graded porous curved beam with defect 1) to demonstrate the fabricated models. Note that we fabricated loading blocks on top of the structures with length and width of 4 mm and 3.3 mm, respectively, to prevent the load cells from contacting other parts of the specimens when undergoing large deformations. The resolution for the 3D printing was set to 100 microns. Because we focus on the in-plane behaviors of the structures, the specimens were printed along the thickness direction to alleviate the influence of possible manufacturing errors (induced by the layerwise printing process) on the in-plane structural properties.

The printing material was ultraviolet photosensitive liquid resin. We used two types of resins with different material properties for manufacturing the MBB beam and curved beam, respectively. To characterize the stress–strain relations of the materials, we performed uniaxial tensile tests according

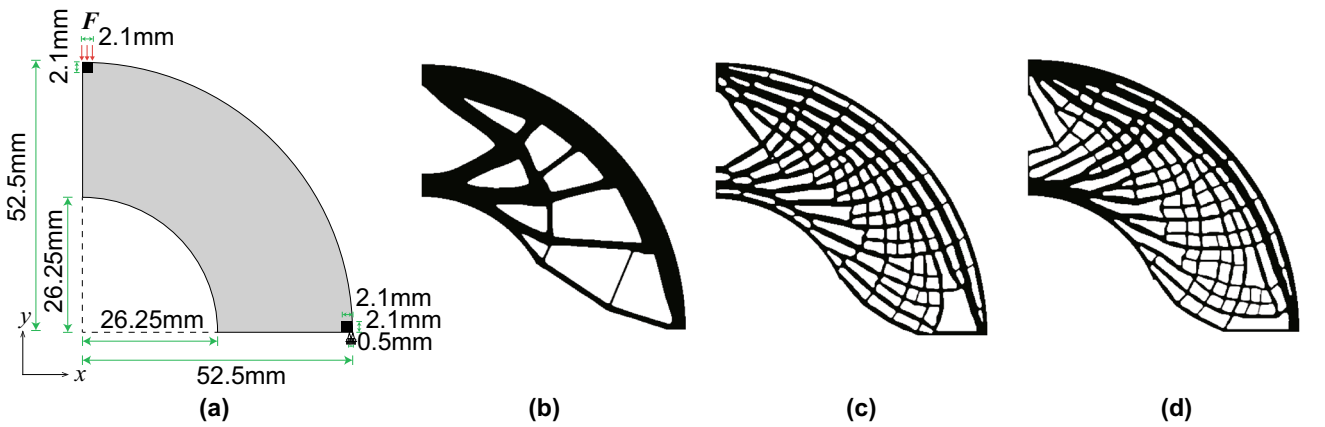


Fig. 2. Topology optimization of curved beam: (a) design domain and boundary conditions, (b) optimized bulk design, (c) optimized one-level porous structure, and (d) optimized graded porous structure.

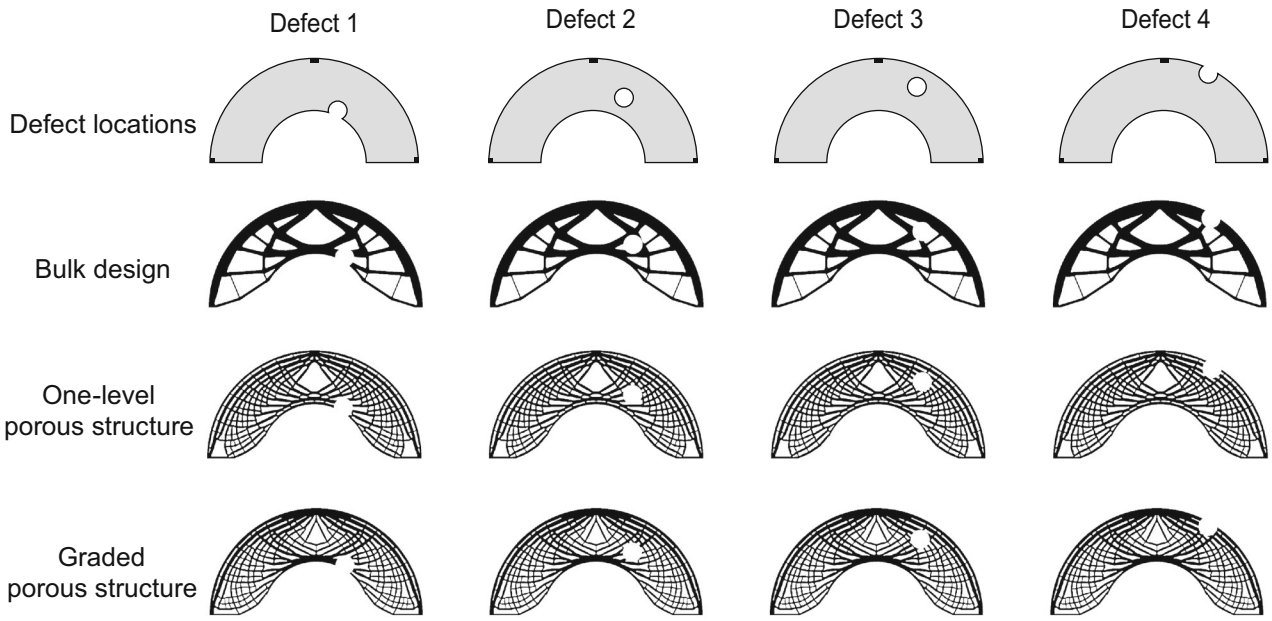


Fig. 3. Optimized curved beam designs with material defects at four different locations.

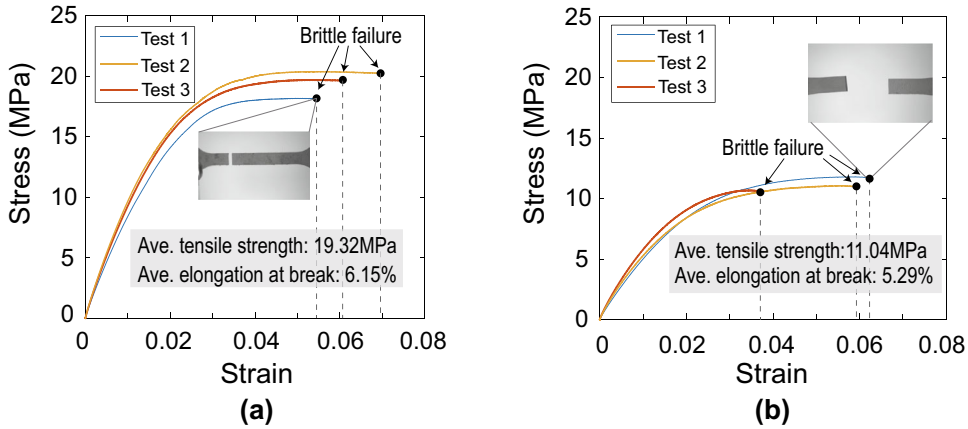


Fig. 4. Experimentally measured stress–strain relations of resin materials for manufacturing (a) MBB beam and (b) curved beam.

to the guidelines of standard ISO 527, which is suitable for characterizing photosensitive resin material.[19] Due to the limitation of the printable size of the 3D printer, we selected a small specimen type (ISO527-2-1BA) for the tests with an overall length of 75 mm, width at the ends of 10 mm, and thickness of 2 mm. An extensometer was used for measuring strain. The displacement-controlled loading rate was 1 mm/min. Three tests were conducted for each material, and the measured results (i.e., stress–strain curves and averaged properties) are shown in Fig. 4. Note that all the material specimens exhibited brittle failure with fracture strain of 6.15% and 5.29%, respectively.

The experiments consisted of two parts: the failure test for the MBB beams and the fail-safe test for the curved beams with material defects. Both of the experiments were carried out using a uniaxial electromechanical loading frame based on the three-point bending test setup as shown in Fig. 5a, b. Displacement-controlled loading was applied, and the loading rates of the failure test and fail-safe test were set to 2 mm/min and 1 mm/min, respectively. The magnitudes of the load cells for the two tests were correspondingly set to 5 kN and 1 kN. For the failure tests, because the structural deformation is large, roller supports can easily cause the specimens to fall from the supports without measuring sufficiently long force–displacement curves. Thus, we placed the specimens on flat surfaces.[10] For the fail-safe tests, we used roller supports because the structural deformation was not large enough to cause the specimens to fall off. We sampled the measured displacements and forces at a rate of 10 measurements per second. Pictures of the structural deformations were captured along the loading processes, using a high-speed camera.

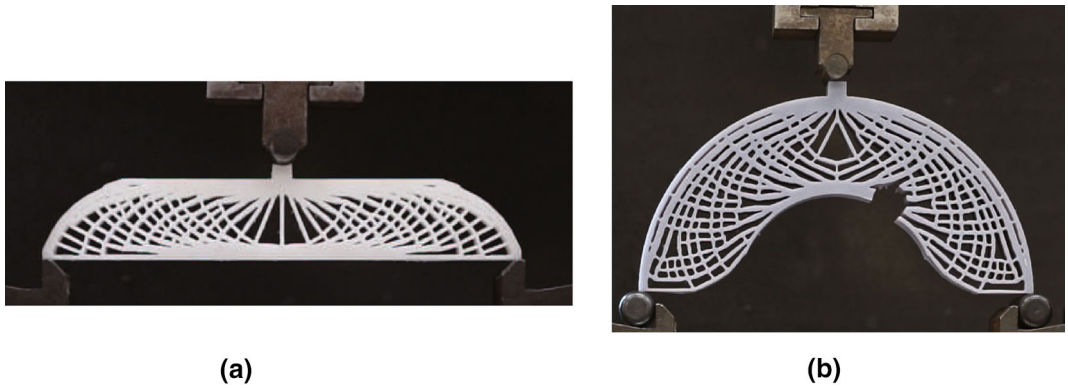


Fig. 5. Demonstrations of 3D-printed models and experiment setups (graded porous structures are used as examples): (a) MBB beam and (b) curved beam with defect 1.

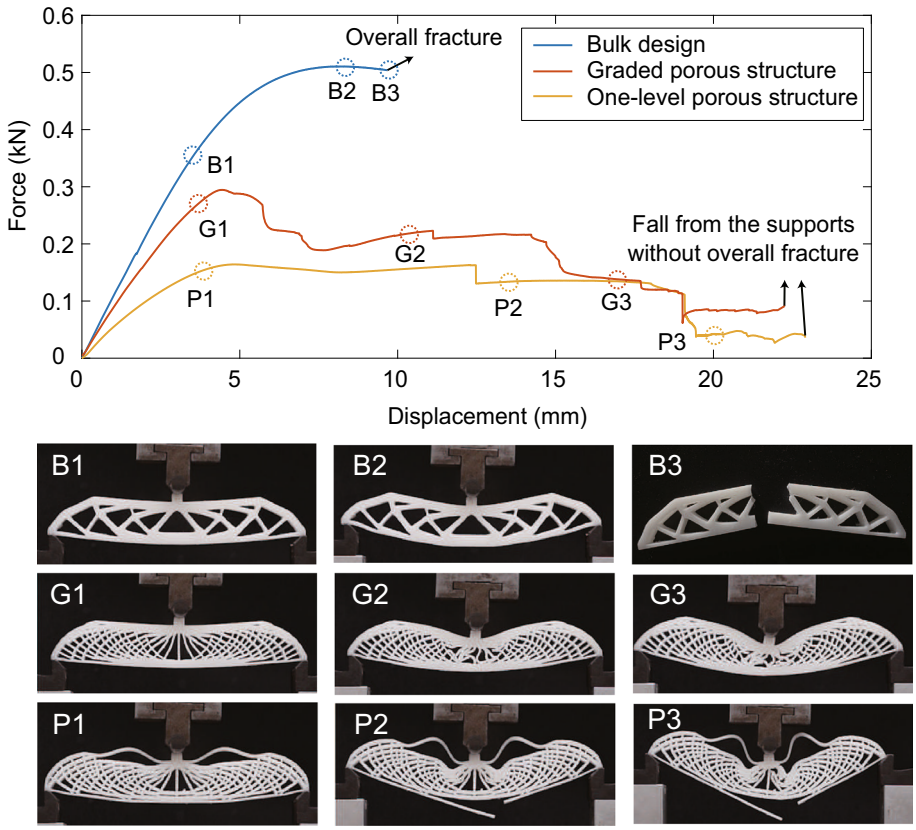


Fig. 6. Force–displacement curves and pictorial demonstrations of failure processes for bulk, one-level porous, and graded porous MBB beams. Note that the overall fracture of the bulk design could not be captured by the camera, and the picture of B3 with an accidentally lost loading block was taken after the experiment. The graded porous structure combines the advantages of the one-level porous structure and the bulk design, achieving a ductile failure mode with a relatively large load capacity and high structural stiffness.

EXPERIMENTAL RESULTS AND DISCUSSION

This section presents and discusses the experimental results for the failure test of the MBB beams and the fail-safe test of the curved beams.

Failure Test: MBB Beam

This test aims to investigate the failure modes and load capacity of the bulk, one-level porous, and graded porous MBB beams. The three types of structures were loaded until structural failure. The force–displacement curves with pictorial demonstrations of the failure processes are shown in Fig. 6. Note that, for the bulk design, the overall

fracture occurred suddenly when the specimen was subjected to displacement of 9.7 mm, while no other local fracture occurred before the overall fracture. Because the bulk design has fewer structural members and less structural redundancy than the porous structures under the stiffness-maximization formulation, the brittle failure mode mainly stems from the brittle material property observed from the measured material constitutive relations in Fig. 4a. Compared with the bulk design, the one-level porous and graded porous structures undergo larger deformations with partial member buckling or breaking, as shown in P2, P3, G2, and G3 of Fig. 6. For the one-level porous and graded porous structures, the large deformations caused the specimens to fall from the supports. We can infer that the overall fractures should happen after 22 mm, which is significantly larger than the value for the bulk design (i.e., 9.7 mm). This phenomenon suggests that the porosity may induce robust load-transferring systems and increase the structural redundancy, thus enabling the ductile failure mode. Even though the material used for manufacturing the one-level porous and graded porous structures is brittle, the fracture at the structural scale is no longer brittle because of the porosity present in the optimized structures.

On the other hand, in terms of the load-carrying capacity among the three types of structures, we find that the bulk design, which possesses the most efficient load-transferring bulk members, sustains the highest load level. According to the initial slope of the force–displacement curve, the bulk design showed the highest stiffness. The one-level porous structure with thinner members sustained the lowest load level and had the lowest stiffness (based on the initial slope of the force–displacement curve). As the design combines both bulk and porosity, the graded porous structure undergoes an intermediate load level and had intermediate stiffness.

Combining the two aspects of structural performance, i.e., failure mode and load capacity (and structural stiffness), we can see a trend that, for the same volume usage, the optimized structure exhibits a more ductile failure mode (which can also be deemed to be a type of structural robustness) at the cost of load capacity and structural stiffness. This finding is aligned with conclusions reached in the robust topology optimization field.[20–21] From the observations of the failure test, the bulk design and one-level porous structure exhibited structural performance of either high load capacity (and structural stiffness) or ductile failure mode, while the graded porous structure achieved a balance between the high stiffness of the bulk design and the robustness of the one-level porous structure, resulting in a structural system with a ductile failure mode and relatively large load capacity and high structural stiffness. Therefore, the failure tests validate that the graded porous structures optimized in the present multiporosity topology optimization framework can provide designers greater freedom to design structures with the combined advantages of structural ductility and relatively high load capacity (and structural stiffness).

Fail-Safe Test: Curved Beam

The fail-safe test aims to compare the structural elastic stiffness of the bulk design, the one-level porous structure, and graded porous structure with respect to different material defects. We loaded the three types of structures until the measured displacement reached 1 mm (i.e., the specimens stayed within the elastic range). The captured force–displacement curves were linearly fit to obtain the equivalent linear stiffness for the comparison in Fig. 7. This figure reveals that, in comparison with the one-level porous and graded porous structures, the stiffness of the bulk design remained at a high level for defects 1–3 but dropped significantly for defect 4. As shown in the inset pictures of the specimens with defect 4 (Fig. 7), two load-transferring paths of the bulk design failed because of defect

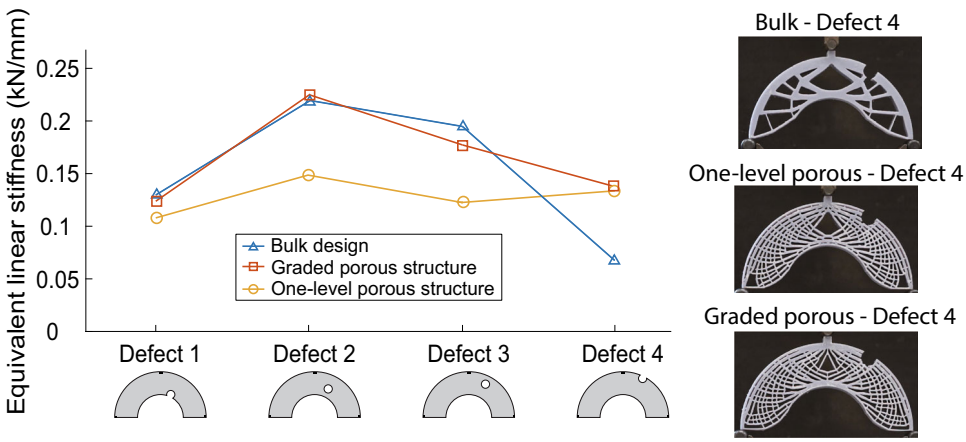


Fig. 7. Experimental fail-safe analysis: comparisons of equivalent linear stiffness for the bulk, one-level porous, and graded porous curved beams (left) and three fabricated beams with defect 4 (right).

4. Therefore, the stiffness reduction of the bulk design with defect 4 is the largest. Note that the fail-safe behavior is influenced by the defect size—with smaller defect size (e.g., smaller than the member size), the stiffness reduction of bulk designs is expected to be less. Compared with the graded porous structure, although the change in the stiffness variation for the one-level porous structure is smaller under various material deficiencies (defects 1–4), the absolute magnitude of the stiffness is lower than the one of the graded porous structure. The experimental observations are consistent with the findings of the numerical fail-safe analysis in Ref. 12. In addition, we note that a large number of computational studies consider structural robustness using rigorous mathematical formulations, namely robust topology optimization (see, e.g., Refs. 20–22). Results of studies in the robust topology optimization field[20–21] have shown that increasing the robustness of designs generally leads to a decrease of the deterministic stiffness, which is aligned with the current findings from the fail-safe tests. In conclusion, the fail-safe test demonstrated that the graded porous structure, balancing the high-stiffness bulk design and highly robust porous structure, showed relatively high elastic load-carrying capacity and robustness with respect to material deficiency.

CONCLUSION

This study experimentally investigated additively manufactured optimized graded porous structures with smooth transitions among multiple levels of porosity and compared various structural characteristics with those of optimized bulk designs and one-level porous structures. We designed the graded porous structures using a novel multiporosity topology optimization framework. The optimized graded porous design achieved an automatic distribution of the multilevel porosity, which combines the design features of both the bulk and one-level porous structures. All the topology-optimized designs were additively manufactured via the MSLA process. We experimentally tested and compared the failure processes and the elastic stiffness under material deficiency for the three types of structures using two examples, i.e., MBB beams and curved beams. The experimental results demonstrated that the additively manufactured graded porous structures, with the optimized combination of bulk members and porous regions, showed a ductile failure mode and robustness against material deficiency with relatively high structural stiffness and load-carrying capacity. Thus, the graded porous designs and the corresponding multiporosity topology optimization design framework could contribute to engineering applications that require both high structural stiffness and large damage tolerance.

Note that the objective in the proposed formulation is to maximize the structural stiffness without mathematically accounting for robustness or fracture in the formulation. Thus, the advantageous structural characteristics such as the ductile failure mode and robustness were experimentally demonstrated in the porous designs but not mathematically guaranteed by the multiporosity topology optimization framework. In terms of the manufacturing process, one limitation is that the layerwise MSLA manufacturing technique could result in directional bias of the structural properties and lead to possible errors in the experiments. One future direction is to consider buckling constraints in the optimization framework to take into account structural buckling. While the presented examples are in 2D for clarity, the presented multiporosity topology optimization framework is general and allows for an extension to 3D with the effective realization of various levels of porosity. Thus, it is promising to experimentally investigate the structural performances of intricate 3D porous structures.

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CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there are no conflicts of interest.

REPLICATION OF RESULTS

Data are available from authors upon request.

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